



Augmented Buyang Huanwu Decoction facilitates axonal regeneration after peripheral nerve transection through the regulation of inflammatory cytokine production

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ABSTRACT

Ethnopharmacological relevance: Herbal formulation Buyang Huanwu Decoction (BYHWD) has been used to treat cardiovascular disorders including cerebral ischemia. Recent studies showed its effects on promoting axonal regeneration after nerve injury. However, compositional reformulation supplemented with herbal components that regulates inflammation may increase its efficacy for nerve repair.

Aim of the study: We prepared a new herbal decoction by adding selected herbal components to BYHWD (augmented BYHWD; ABHD) and investigated the effect of ABHD on the production of inflammatory cytokines and axonal regeneration using an animal model of nerve transection and coaptation (NTC).

Materials and methods: A rat model of NTC was performed on the sciatic nerve. The sciatic nerve and dorsal root ganglion (DRG) were isolated and used for immunofluorescence staining and western blot analysis. DRG tissue was also used to prepare primary neuron culture and the length of neurites was analyzed. Sensorimotor nerve activities were assessed by rotarod and von Frey tests.

Results: Three herbal components that facilitated neurite outgrowth were chosen to formulate ABHD. ABHD administration into the sciatic nerve 1 week or 3 months after NTC facilitated axonal regeneration. Cell division cycle 2 (Cdc2) and brain-derived neurotrophic factor (BDNF) proteins were induced from the reconnected distal portion of the sciatic nerve and the levels were further elevated by in vivo administration of ABHD. Phospho-Erk1/2 level was increased by ABHD treatment as well, implying its role in mediating retrograde transport of BDNF signals into the neuronal cell body. Production of inflammatory cytokines IL-1 β and TNF- α was induced in the reconnected nerve but attenuated by ABHD treatment. Behavioral tests revealed that ABHD treatment improved functional recovery of sensorimotor activities.

Conclusions: A newly formulated ABHD is effective at regulating the production of inflammatory cytokines and promoting axonal regeneration after nerve transection and may be considered to develop therapeutic strategies for peripheral nerve injury disorders.

1. Introduction

Axons in the peripheral nerve can regenerate after injury, regrow to the original target and lead to functional recovery. Here, neurons having axonal injury have intrinsic capacity to regenerate via the interaction with lesion-induced molecules and subsequently cell body response (Mahar and Cavalli, 2018; Chen et al., 2007). In addition, environmental factors around the injury site are supportive of promoting axonal elongation and correct targeting. Schwann cells in the injured nerve are

known to dedifferentiate following the dissociation from the axons, proliferate, distally migrate, and interact again with regrowing axons to support axonal elongation (Napoli et al., 2012; Namgung, 2014).

Unlike the experimental animals, however, axonal regeneration after peripheral nerve injury in humans is much limited, and consequent functional recovery is far from perfect (Buchthal and Kühl, 1979). This is partly due to the fact that a distance from the injury site to the axonal target area is much longer in human than rodents. In human, potentially positive environmental factors for axon regeneration including Schwann

Abbreviations: ABHD, Augmented Buyang Huanwu decoction; BYHWD, Buyang Huanwu decoction; NTC, Nerve transection and coaptation; Cdc2, cell division cycle 2; DRG, dorsal root ganglion; (GAP)-43, growth-associated protein.

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cells are dispersed all through the distal nerve areas, and those particularly localized in the distant portion of the nerve would have much less chance to interact with regenerating axons and thus, they may be functionally inactive in mediating axonal regeneration (Höke, 2006). In order to induce the timely interaction, a strategy to increase the rate of axonal regeneration was accomplished by using transgenic animals that overexpress heat shock protein 27 (Ma et al., 2011). Alternatively, activation of Schwann cells in the denervated nerve by electrical stimulation or nerve graft may support axonal regeneration (Boyd and Gordon, 2003; Gordon, 2016). Taken together, understanding the mechanistic basis on the interaction between regenerating axons and distal environmental factors is of great importance to develop therapeutic strategies of nerve injury disorders in human.

BYHWD has been formulated to supplement the vital energy and facilitate the flow of blood, and several reports have demonstrated that BYHWD was effective at reducing ischemic injury in several organs (Li et al., 2014; Yang et al., 2015; Pan et al., 2017) and ameliorating neural function by protecting brain cells and peripheral Schwann cells from injury (Yang et al., 2011; Li et al., 2003). In order to further improve its efficacy on regulation of cardiovascular and neurological functions, modified BYHWD was developed and its clinical effects in terms of neurological indices were reported in patients with the early stroke, cerebral infarction, or cervical spinal canal stenosis (Wang et al., 2013; Wan and Hou, 2017; Lu et al., 2019). Also, studies using in experimental animals showed that the administration of modified BYHWD supplemented with *Achyranthes Radix*, *Salviae Miltiorrhizae Radix*, *Cinnamomi Ramulus*, *Polygalae Radix*, and *Acori Graminei Rhizoma* improved the recovery of neuronal tissues from cerebral damage after cerebral artery occlusion and reperfusion (Choi et al., 2011; Oh et al., 2015), and altered formulation of modified BYHWD which was supplemented with *Chaenomeles speciosa*, *Chinemys reevesii*, *Rehmannia glutinosa*, *Eucommia ulmoides*, *Achyranthes bidentata*, *Panax notoginseng*, *Ligustrum lucidus*, *Cynomorium songaricum*, and *Dipsacus asper* facilitated regeneration of disconnected peripheral nerve through silicone rubber tubes (Chen et al., 2001). Therefore, these studies imply that the pharmacological effects of BYHWD on neurological and cardiovascular functions may be improved by modifying herbal composition.

Previously, we demonstrated that BYHWD facilitated axonal regeneration in the sciatic nerve in which the nerve injury was given by crush injury or transection and coaptation (Chang et al., 2016; Kim and Namgung, 2018), and more recently, further showed that modified BYHWD in which *Lumbricus*, a genus of earthworm, was omitted was more effective at regulating inflammation (Ryu et al., 2020). Given that some of therapeutic agents aim to regulate both neuropathic pain and nerve repair (Chuang et al., 2018; Cobianchi et al., 2017; Ding et al., 2017), we designed to examine the effects of newly formulated decoctions on axonal regeneration. Specifically, we surveyed and selected herbal drugs such as *Achyranthes bidentata* Blume, *Phlomis umbrosa* Turcz., *Aralia continentalis* Kitag., *Salvia miltiorrhiza* Bunge, *Eucommia ulmoides* Oliv., *Clematis mandschurica* Max., *Angelica grosseserrata* Maxim, *Cinnamomum verum*, and *Lonicera japonica* Thunb., and explored their activity on peripheral nerve regeneration. Pathophysiological activities of selected herbal drugs that are related to the regulation of inflammation and nociception, neuroprotection, and nerve repair, as reported in several lines of experimental studies, are listed in Table 1. Yet, their combinatorial activities on the regulation of peripheral nerve repair in association with the efficacy of BYHWD have not been investigated. Here, we report that, using animal model of nerve transection and coaptation (NTC), an augmented BYHWD in which herbal drugs *Achyranthes bidentata*, *Phlomis umbrosa*, and *Aralia continentalis* were supplemented was effective at regulating pathological responses in the injured nerve and thus facilitating axonal regeneration leading to functional recovery.

Table 1

Information of Herbal drugs used in this study.

^a Botanical name	Common herbal name (or English name)	Specimen number (Production area)	% dry weight in ABHD
<i>Astragalus membranaceus</i> (Fisch.) Bunge ^b	<i>Radix Astragali</i> (Mongolian milkvetch)	KNP0011 (Jeongseon, Korea)	77.2
<i>Angelica sinensis</i> (Oliv.) Diels ^b	<i>Radix Angelica Sinensis</i> (Astragalus)	KNP0008 (Jeongseon, Korea)	3.9
<i>Paeonia lactiflora</i> Pall. ^b	<i>Radix Paeoniae Rubra</i> (Red Paeony Root)	KNP0085 (Younghae, Korea)	2.9
<i>Ligusticum chuanxiong</i> Hort. ^b	<i>Rhizoma Ligustici Chuanxiong</i> (Szechuan Lovage Rhizome)	KNP0028 (Hebei, China)	1.9
<i>Carthami flos</i> ^b	<i>Carthamus tinctorius</i> L. (Safflower)	KNP0020 (Xingang, Taiwan)	1.9
<i>Prunus persica</i> (L.) Batsch ^b	<i>Semen Persicae</i> (Peach seed)	KNP0087 (Hebei, China)	1.9
<i>Achyranthes bidentata</i> Blume ^c	<i>Achyranthes bidentata</i> (ox knee)	KNP0034 (Andong, Korea)	2.6
<i>Phlomis umbrosa</i> Turcz. ^c	<i>Phlomis umbrosa</i>	KNP0031 (Sichuan, China)	5.1
<i>Aralia continentalis</i> Kitag. ^c	<i>Aralia continentalis</i>	KNP0033 (Muju, Korea)	2.6
<i>Salvia miltiorrhiza</i> Bunge ^d	<i>Salvia miltiorrhiza</i> root and rhizome	KNP0045 (Hebei, China)	–
<i>Eucommia ulmoides</i> Oliv. ^d	<i>Eucommia ulmoides</i>	KNP0055 (Kyongbuk, Korea)	–
<i>Clematis mandschurica</i> Max. ^d	<i>Clematis mandschurica</i>	KNP0080 (Hebei, China)	–
<i>Angelica grosseserrata</i> Maxim. ^d	<i>Angelica koreana</i> L.	KNP0076 (Gangwon-do, Korea)	–
<i>Cinnamomum verum</i> ^d	Cinnamon	KNP0047 (Yen Bai, Vietnam)	–
<i>Lonicera japonica</i> Thunb. ^d	<i>Lonicera japonica</i> (Japanese honeysuckle)	KNP0072 (Ulsan, Korea)	–

^a Numbers in parenthesis.^b , components of BYHWD.^c , selected components for ABHD.^d Non-selected components.

2. Materials and Methods

2.1. Experimental animals

Sprague-Dawley rats (male, 200–250 g) and mice (albino ICR; 20–25 g) were obtained from Samtako (Seoul, Korea). Animal rooms were maintained at 22°C, 60% humidity, and a 12 h light and 12 h dark cycle. Animals were allowed to eat commercial chow (Samyang Co., Korea) and drink water ad libitum. Animal care and all experimental procedures were in accordance with the NIH guide for the care and use of laboratory animals and approved by the Committee on Use of Live Animals for Teaching and Research at Daejeon University (Protocol number: DJUAR2019-028; Daejeon, Korea).

2.2. Drugs

Augmented BYHWD (ABHD) is composed of 6 herbs of BYHWD but *Lumbricus* (; *Radix Astragali*, *Radix Angelica Sinensis*, *Radix Paeoniae Rubra*, *Rhizoma Ligustici*, *Chuanxiong*, *Carthamus tinctorius* L. (Chang et al., 2016) and 3 newly selected components (*Achyranthes bidentata*, *Phlomis umbrosa*, and *Aralia continentalis*). Information of the herbal ingredients that were examined in the present study is summarized in Table 2. All herbal drugs were purchased from Omniherb (Yeongcheon, Korea) and were deposited in the Herbarium of the Department of

Table 2

Major pathophysiological effects of herbal drugs used in this study.

Botanical name	Major pathophysiological activities	References
<i>Achyranthes bidentata</i> Blume	anti-inflammatory, anti-apoptotic	Xu et al. (2017)
<i>Phlomis umbrosa</i> Turcz.	antinociceptive, anti-inflammatory,	Shang et al. (2011)
<i>Aralia continentalis</i> Kitag	anti-inflammatory in traumatic brain	Lee et al. (2019)
<i>Salvia miltiorrhiza</i> Bunge	anti-oxidative, promoting nerve regeneration, anti-cancer	Li et al. (2017); Lin et al. (2019); Koo et al. (2004)
<i>Eucommia ulmoides</i> Oliv.	antioxidative, erectile function, wide-range of pharmacological activities	Li et al. (2016); Fu et al. (2019); He et al. (2014)
<i>Clematis mandshurica</i> Max.	anti-inflammatory	Dilshara et al. (2014)
<i>Angelica grosseserrata</i> Maxim	hypopigmenting activity	Roh et al. (2011)
<i>Cinnamomum verum</i>	neuroprotective, anti-cancer	Khasnavis and Pahan (2014); Sadeghi et al. (2019)
<i>Lonicera japonica</i> Thunb.	anti-inflammatory, analgesic	Ryu et al. (2010)

Korean Medicine, Daejeon University (Daejeon, Korea). Extraction procedure was essentially the same as described previously (Chang et al., 2016; Kim et al., 2018). Briefly, individual, dried herbal drugs were chopped into small pieces, soaked into distilled water and boiled for 2 h. After filtration three times using Whatman filter papers (Grade 1, Whatman Inc., Clifton, USA), the extracts were subjected to freeze drying. The yields for individual drugs for initial raw materials (dry weight) were variable, showing about 27% for *Salvia miltiorrhiza* and *Achyranthes bidentata*, 5–10% for *Angelica grosseserrata*, *Clematis mandshurica*, *Eucommia ulmoides*, *Phlomis umbrosa*, and *Aralia continentalis*, and about 4% for *Cinnamomum verum* and *Lonicera japonica*. The products from freeze drying were dissolved in saline solution (1.0 g/ml) and stored at -20°C .

2.3. Nerve transection and coaptation

Rats were anesthetized by injecting ip with a mixture of ketamine (80 mg/kg) and xylazine (5 mg/kg). Sciatic nerve was exposed on the middle of the hindlimb, cut, and both proximal and distal ends of the nerve were fixed on nearby muscle tissues as described previously (Kim and Namgung, 2018). After suture and recovery from anesthesia, animals were returned to animal room, and 1 week or 3 months later, they were subjected to coaptation surgery. The nerve ends were carefully separated from the attached tissue and reconnected by suturing 3 locations of aligned nerves using 7–0 nylon (Ethicon, Somerville, USA). Animals were recovered from the surgery and administered with herbal drugs (400 mg/kg) or an equivalent volume of saline vehicle on a daily basis for 14 days. Our previous report showed that the oral administration of BYHWD with a dose of 400 mg/kg was optimal for inducing regenerative responses in the sciatic nerve after injury (Chang et al., 2016) and thus the same dose was orally applied in the present investigation. Animals were then sacrificed for biochemical and immunohistochemical analysis of nerve tissues.

2.4. Western blot analysis

Sciatic nerve and DRG at lumbar levels 4–6 were isolated from rats and used to prepare protein lysates as described in our previous report (Chang et al., 2012a). Protein samples (15 μg each) were used for SDS-polyacrylamide gel electrophoresis (SDS-PAGE) and western blot analysis. Antibodies used were GAP-43 (Abcam, Cambridge, UK; Rabbit

1000:1) anti-Cdc2 (Santa Cruz Biotech; mouse, 1000:1), anti-IL-1 β (Abcam; Rabbit 1000:1), anti-TNF- α (Abcam; Rabbit 1000:1), anti-BDNF (Abcam; Rabbit 1000:1), anti-TrkB (Abcam; Rabbit 1000:1), anti-phospho-Erk1/2 (Cell Signaling Technology, Beverly, USA; Rabbit 1000:1), and anti-Actin (Sigma; mouse 20000:1) primary antibodies and goat anti-mouse IgG (Genetex, Irvine, USA; 1000:1) or goat anti-rabbit IgG (Santa Cruz Biotechnology, Dallas, USA; 1000:1) secondary antibodies. Intensity of protein bands was quantitated by using the i-Solution software (Image & Microscope Technology, Daejeon, Korea).

2.5. Immunofluorescence staining

The sciatic nerve and DRG tissues were dissected from rats, rapidly frozen and used for the preparation of longitudinal section (20 μm) by using cryostat (Leica, Wetzlar, Germany). Pretreatment, blocking, and primary and secondary antibody reactions were performed as described previously (Chang et al., 2016). Antibodies used were anti-NF-200 (monoclonal, Sigma, St. Louis, USA; 800:1), anti-SCG10 (polyclonal, NovusBio, Centennial, USA; 800:1), anti-phospho-Erk1/2 (polyclonal, Cell Signaling Technology; 800:1), anti-TNF- α (polyclonal, Abcam; 800:1), anti-IL-1 β (polyclonal, Abcam; 400:1), anti-TrkB (polyclonal, Abcam; 300:1) primary antibodies and fluorescein goat anti-mouse and rhodamine goat anti-rabbit (Life Technologies; 800:1) secondary antibodies. Sections were also reacted with Hoechst33258 (Sigma, 2.5 $\mu\text{g}/\text{ml}$) to visualize nuclei. Fluorescence images were observed by using fluorescence microscope (Nikon, Kona, Japan), captured and analyzed by Adobe Photo Program.

2.6. Culture of DRG neurons

DRG at lumbar 4–6 levels were isolated from intact rats or from rats whose sciatic nerve had been transected for 7 days. Primary culture of DRG sensory neurons was performed as described previously (Chang et al., 2012a). Briefly, cells (1.5×10^2) were plated on 12 mm coverslips coated with laminin (0.02 mg/ml, Collaborative Research, Bedford, USA) and polyornithine (0.01%, Sigma). Cells were treated with herbal drugs (0.1 mg/ml) or with equivalent volume of saline for 48 h prior to harvest for fixation. In our previous study examining the effects of BYHWD on neurite outgrowth of primary DRG neurons, a treatment with 0.5 mg/ml of BYHWD was twice as effective as a treatment of 0.1 mg/kg (Chang et al., 2016). However, we used 0.1 mg/ml in the present study because this lower dose, possibly causing less pharmacological side effects than the higher dose, appeared to be high enough to compare the relative efficacy of enhancing neurite outgrowth among individual herbal drugs. Cells on the coverslips were used for immunofluorescence staining, and the immunostaining procedure was essentially same as describe above. Length of neurites of individual neurons was measured by using the i-Solution software (Image & Microscope Technology, Daejeon, Korea). Fifteen to twenty neurons were randomly selected from the microscopic fields and the average length of neurites was determined. Mean neuritic length was obtained from 4 independent experiments and compared among experimental groups.

2.7. Behavioral tests

For behavioral tests, mice (albino ICR 25 g male) were subjected to NTC surgery in the sciatic nerve. Surgery procedures for mice were similar to rats except that the cut nerves were reconnected using 9–0 nylon (Ethicon). Sciatic nerves were transected and coapted 7 days after. Animals were then randomly assigned into saline-treated (NTC + SAL) and ABHD-treated (NTC + ABHD) groups, and ABHD (400 mg/kg) and equivalent volume of saline were administered daily for 14 days. Together with untreated control mice, NTC + SAL and NTC + ABHD groups of animals were used for rotarod test and also subjected to von Frey test 12 h later. Animals were first adjusted for 5 min to the equipment by placing on the non-rotating rod. Then, the animals were

tested to determine the time maintained on the rod which was set to 2–20 of rpm and the mean values were determined after 3 different measurements with 1 h interval between each test. Von Frey test were performed as has been described in our previous report (Kim et al., 2018). Briefly, animals were adjusted to the von Frey equipment by placing them on the equipment for 15 min and the frequency of withdrawal reflex of the hind leg was measured by stimulating the foot 20 times. Mean values of withdrawal reflex were determined by measuring three time with a 30 min interval.

2.8. Statistical analysis

Data were presented as mean $\pm$ standard deviation (SD). The mean values of the data among experimental groups were compared using one-way ANOVA and Dunnett's test for multiple comparisons (Graph-Pad Prism 7.00). Two-way ANOVA with Sidak multiple comparison test was used to compare two groups with two independent variables. Statistically significant differences were set at $p < 0.05$.

3. Results

We have begun to examine 9 individual herbal drugs (underlined bold text in Table 1) that were selected based on their activities for the regulation of inflammation, nociception, neurodegeneration and nerve injury (see Table 2). We prepared primary DRG neurons and analyzed neurite outgrowth after the treatment of candidate herbal drugs to formulate ABHD. Our initial analysis showed no significant changes in neurite outgrowth by adding candidate drugs compared with control group (data not shown). Having noticed, however, that thin, but highly elongated neurites were detected particularly in drug-treated groups, we re-analyzed the neurites whose lengths were longer than 500 μm . Cell groups treated with herbal drugs *Achyranthes bidentate* (AB), *Phlomis umbrosa* (PU), and *Aralia continentalis* (AC) showed significant increases in neurite length compared with saline control (Fig. 1). In a previous report, AB was added 2.4% (wt/wt) as one of the components of modified BYHWD (Chen et al., 2001). PU and a constituent of BYHWD *Radix Angelica* were used with a similar proportion in a clinical study evaluating their effects on the menopausal symptoms (Chang et al., 2012b), and AC is added 50% the amount of PU in Mangeum-tang, one of the herbal formulation in traditional Korean medicine (Choi et al., 2003). Based on the information as mentioned above and also considering some compositional variations in other studies, we added AB, PU, and AC (2.6%, 5.1%, and 2.6% in wt/wt respectively) to the Lumbricus-omitted BYHWD, formulated an augmented BYHWD (ABHD) (Table 1), and used it for the rest of current investigation.

To investigate the effects of ABHD on the regenerative responses of

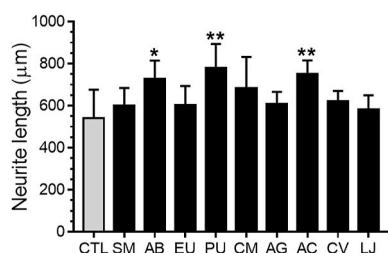


Fig. 1. Comparison of neurite outgrowth of DRG neurons treated with herbal drugs. DRG neurons were prepared from rats given transection injury on the sciatic nerve for 7 days and drugs (0.1 mg/ml) were treated for 48 h prior to fixation for immunofluorescence staining. N (number of experiments) = 4. Error bars (SD). * $p < 0.05$, ** $p < 0.01$ vs. CTL (One-way ANOVA). Abbreviations for individual herbal drugs: *Salvia miltiorrhiza* (SM), *Achyranthes bidentate* (AB), *Eucommia ulmoides* (EU), *Phlomis umbrosa* (PU), *Clematis mandshurica* (CM), *Angelica grosseserrata*. (AG), *Aralia continentalis* (AC), *Cinnamomum verum* (CV), *Lonicera japonica* (LJ).

sciatic nerve axons in NTC animals, individual axons in the longitudinal nerve sections were visualized by immunofluorescence staining. In one set of experiment, the sciatic nerve was transected for 1 week and coapted for 2 weeks (1W-NTC) during which ABHD or saline vehicle was administered (Fig. 2A). Immunofluorescence staining of NF-200 showed that axonal regeneration in the distal stump was clearly improved in ABHD-treated group compared with the saline-treated control (Fig. 2B). Quantification analysis revealed that axonal elongation was similar between saline- and ABHD-treated groups at the distal area close to the transection area, but was improved at farther distal portion of the nerve in ABHD-treated group compared with the saline control, showing significant difference between two groups ($p = 0.0053$, Two-way ANOVA) (Fig. 2C). To investigate the effects of long-term denervation on axonal regeneration, we reconnected the nerve 3 month after transection and treated with drugs for subsequent 2 weeks (3M-NTC in Fig. 2A). Here, we conducted immunofluorescence staining of nerve sections using SCG-10 antibody. Since SCG-10 protein is rapidly degraded in the degenerating axons, immunofluorescence analysis of SCG10 as a marker of regenerating axons can minimize the false labeling of degenerating axons (Shin et al., 2014). It was noted that the extent of axonal regrowth into the distal nerve was drastically reduced in saline-treated control. Then, axonal regeneration was remarkably enhanced by ABHD treatment (Fig. 2D). Statistical comparisons between two groups showed significant difference between NTC + ABHD and NTC + SAL groups ($p = 0.032$, Two-way ANOVA; Fig. 2E).

To examine molecular factors that might be involved in the process of ABHD-mediated axonal regeneration, we investigated axonal growth-associated protein (GAP)-43 and Cdc2 proteins as intrinsic and extrinsic markers of axonal regeneration, respectively (Caroni, 1997; Han et al., 2007; Chang et al., 2012a). GAP-43 was clearly induced in the sciatic nerve in 1W-NTC animals and showed a slight increase by ABHD treatment compared to saline-treated group ($p = 0.9$, One-way ANOVA; Fig. 3A upper and lower panels). To further examine the effects of ABHD on GAP-43 expression and neurite outgrowth, we prepared cultured DRG neurons from rats that had undergone preconditioning transection injury (labeled pINJ in Fig. 3B) for 7 days or from rats with no preconditioning injury. Neurite outgrowth and GAP-43 signal intensity in the preinjury groups were obviously increased compared with those in the control group (Fig. 3B and C). However, overall GAP-43 signal intensity was similar between saline- and ABHD-treated cells except that some intense signals were detected particularly at the neuritic tips of ABHD-treated cells (enlarged in Fig. 3B). There was a slight increase in neurite outgrowth of cells treated with ABHD compared to saline group, failing to show statistical difference ($p = 0.06$, One-way ANOVA; Fig. 3C). Cdc2 protein, known to be induced in Schwann cells and involved in facilitating axonal regeneration after peripheral nerve injury (Han et al., 2007), was strongly induced in the sciatic nerve in NTC animals, and its level was further upregulated by ABHD treatment, showing a significant increase in band intensity (Fig. 3D upper and lower panels).

Previous studies have shown that BDNF is produced and secreted from Schwann cells after peripheral nerve injury and interacts with neuronal receptors such as TrkB and P75 (Richner et al., 2014). TrkB, along with an entrapped BDNF, is retrogradely transported into the cell body and induces cell body response including gene expression (Segal and Greenberg, 2006). Here, BDNF was induced in the sciatic nerve after NTC, and the level further elevated by ABHD treatment (Fig. 4A upper and lower panels). Unlike BDNF, TrkB was produced from untreated control and remained unchanged after NTC and ABHD treatment (Fig. 4B upper and lower panels). Interestingly, TrkB level in the DRG was clearly increased after ABHD treatment in NTC animals (Fig. 4C upper and lower panels). We also investigated whether the activation of Erk1/2 is related to axonal regeneration after NTC. Phospho-Erk1/2 level in the sciatic nerve was not changed after NTC and then elevated by ABHD treatment (Fig. 4D upper and lower panels). In the DRG, moderate level of phospho-Erk1/2 was also observed in the control

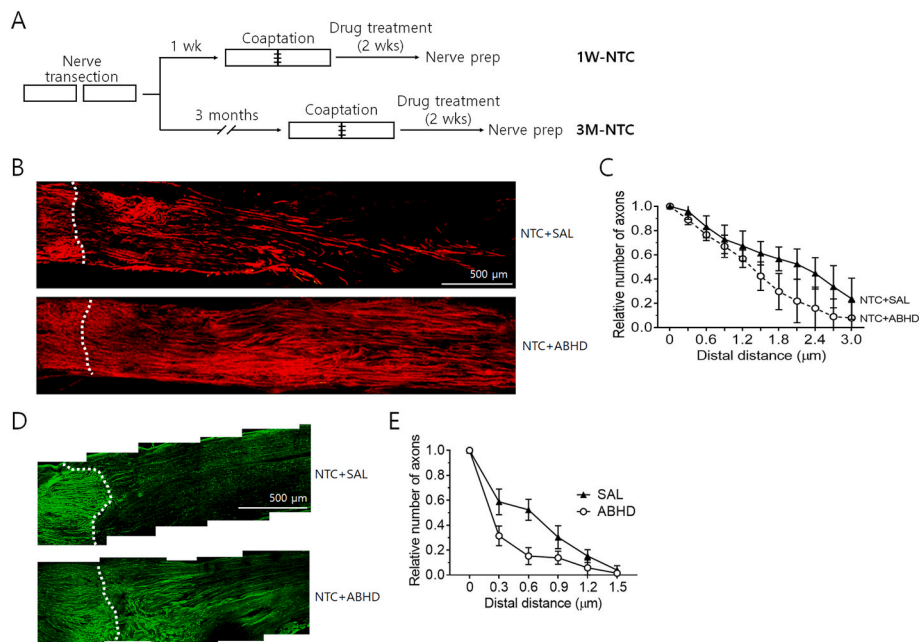


Fig. 2. Enhancement of axonal regeneration by ABHD administration in NTC animals. Longitudinal nerve sections from NTC + SAL and NTC + ABHD groups of animals were used for immunofluorescence staining to visualize regenerating axons. (A) A schematic illustrating a time course of nerve surgery and ABHD or saline treatment in animal groups of 1W-NTC and 3M-NTC. In 1W-NTC group, transected nerve was coapted 1 week later and treated with ABHD or saline for 2 weeks, whereas in 3M-NTC group, the nerve was coapted 3 months after the transection and treated with ABHD for 2 weeks. (B) Representative images of NF-200-stained longitudinal nerve sections in 1W-NTC animals. (C) Quantification of NF-200-stained regenerating axons at different distances from the injury site (marked dotted lines in B) in 1W-NTC animals. $p < 0.01$ ($F(1, 6) = 18.25$, Two-way ANOVA, $n = 7$ nerves per group). (D) Representative images of SCG10-stained longitudinal nerve sections in 3M-NTC animals. (E) Quantification of SCG10-stained regenerating axons at different distances from the injury site (marked dotted lines in D) in 3M-NTC animals. $p < 0.05$ ($F(1, 3) = 14.4$, Two-way ANOVA, $n = 4$ nerves per group). In (C) and (E), four to five nonconsecutive nerve sections per each nerve were used to count the number of axons at different distance from the injury site.

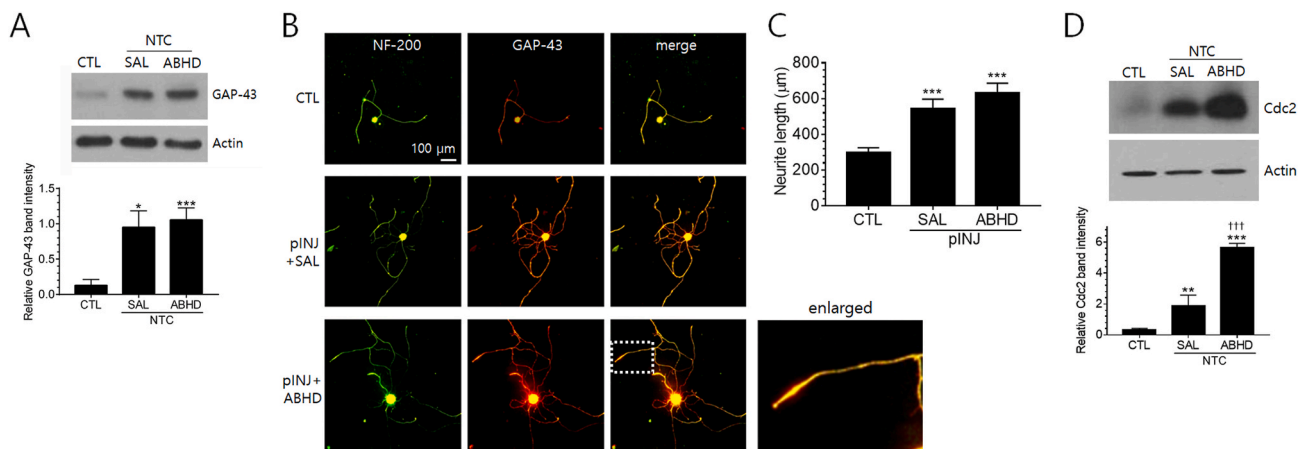


Fig. 3. Effects of NTC and ABHD administration on the production of GAP-43 and Cdc2 proteins. In (A) and (D), proteins extracted from the sciatic nerves of different groups of animals were used for western blotting for GAP-43 (A) and Cdc2 (D). In (A) and (D), western blot images (upper) are the representatives from four independent experiments, and quantifications of GAP-43 and Cdc2 band intensity relative to actin are plotted lower panels in (A) and (D), respectively. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. CTL; † $p < 0.05$ vs. NTC + SAL. Western blottings for actin as sample loading control. (B) Immunofluorescence staining of primary DRG neurons for NF-200 and GAP-43. DRG were prepared from intact animals (CTL) or from animals given transection injury for 7 days (preinjury; pINJ). Cells prepared from pINJ were treated with saline (pINJ + SAL) or ABHD (pINJ + ABHD) for 48 h prior to fixation for double immunofluorescence staining with anti-NF-200 and anti-GAP-43 antibodies. (C) Quantification of neurite outgrowth of cultured DRG neurons. *** $p < 0.05$ (One-way ANOVA, $n = 4$).

animals, and its level was upregulated in NTC animals and further increased by ABHD (Fig. 4E upper and lower panels). Immunofluorescence staining of sciatic nerve sections prepared from ABHD-treated NTC animals showed that TrkB and phospho-Erk1/2 signals were highly colocalized (Fig. 4F). In the DRG of ABHD-treated NTC animals, phospho-Erk1/2 signals were clearly seen in NF-200-labeled sensory neurons and largely colocalized with TrkB signals (Fig. 4G). Taken together, these data suggest that retrograde transport of BDNF/TrkB and phospho-Erk1/2 signals that are induced from injured axons after NTC may be facilitated by ABHD.

An NTC surgery can induce pathological inflammatory responses which may affect the process of nerve repair. Thus, we analyzed the production of proinflammatory cytokines TNF- α and IL-1 β in the sciatic nerve and DRG tissues and examined the regulatory effects of ABHD in NTC animals. In the sciatic nerve, production of TNF- α protein, which

was induced in the sciatic nerve after NTC, was significantly decreased by ABHD (Fig. 5A upper and lower panels). Immunofluorescence staining showed a clear induction of TNF- α in the distal portion of the nerve after NTC, which was greatly suppressed by ABHD treatment (Fig. 5B). TNF- α signals in saline-treated NTC group were rarely colocalized with NF-200-stained axons, suggesting that TNF- α may be mostly produced from non-neuronal cells such as Schwann cells and infiltrated macrophages. Consistent with western data, overall signal intensity of TNF- α was markedly decreased by ABHD treatment. Similarly, TNF- α signals were clearly induced in the DRG of NTC animals and suppressed by ABHD treatment (Fig. 5C). TNF- α signals, which were observed well in NTC + SAL group of animals, were colocalized with DRG neurons labeled with NF-200, showing more intense signals in the center where NF-200 signals were relatively weak possibly in the nucleus area (marked arrows in the enlarged images in Fig. 5C).

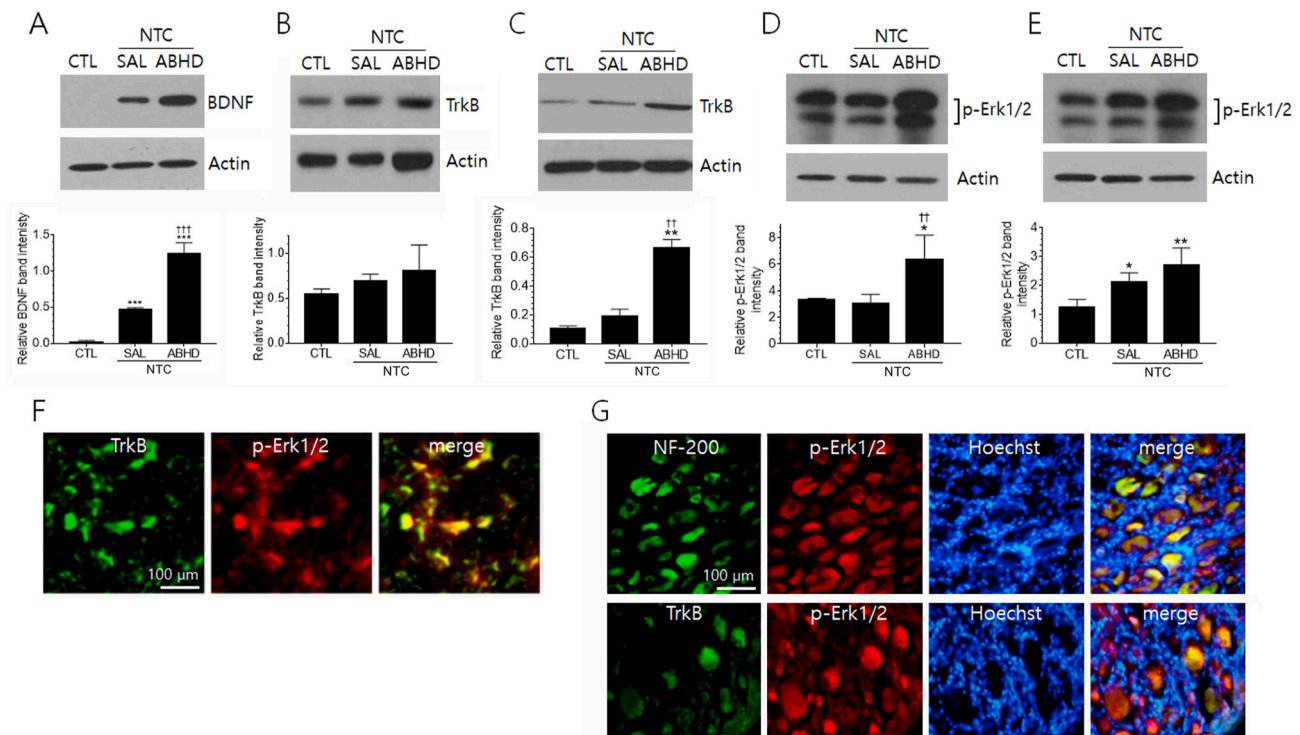


Fig. 4. Regulation on the production of BDNF, TrkB and phospho-Erk1/2 in the sciatic nerve and DRG by ABHD treatment in NTC animals. (A–E) Western blot analysis for the proteins as indicated in the figure. Protein samples in (A), (B) and (D) were prepared from the sciatic nerve and those in (C) and (E) from the DRG. Western blot images (upper in A–E) are the representatives from four independent experiments, and quantifications of protein band intensity relative to actin are plotted in lower panels. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. CTL; †† $p < 0.01$, ††† $p < 0.001$ vs. NTC + SAL. Western blottings for actin as sample loading control. (F, G) Transverse sciatic nerve (F) and DRG tissue (G) sections were prepared from ABHD + NTC group of animals and used for double immunofluorescence staining for NF-200, TrkB and phospho-Erk1/2 proteins. Individual nuclei in the DRG were visualized by Hoechst staining ...

We also investigated the production of IL-1 β in NTC animals after ABHD administration. IL-1 β was highly induced in the sciatic nerve after NTC and significantly decreased by ABHD treatment (Fig. 6A upper and lower panels). Immunofluorescence analysis showed that IL-1 β signals were clearly observed not only in the sciatic nerve but also in the DRG of NTC + SAL group of animals (Fig. 6B and C). Similar to the signaling pattern of TNF- α , most of the IL-1 β signals were not detected from NF-200-labeled axons in the sciatic nerve and the intense signals in DRG neurons were centrally localized in neuronal cell body (Arrows in Fig. 6C). IL-1 β signals in the sciatic nerve and DRG were greatly diminished by ABHD treatment.

To determine whether improved axonal regeneration by ABHD in NTC animals was linked to behavioral consequences, we analyzed sensorimotor activity of the hind legs. In NTC group of animals, the latency to fall off the rotarod was greatly decreased compared with control group, and ABHD administration significantly improved the responses (Fig. 7A). When we measured behaviors of pain sensitivity by von Frey test, a group of animals given NTC showed almost 4 fold-increases in pain sensitivity (Fig. 7B). Then, the treatment of ABHD significantly attenuated pain sensitivity compared with ABHD + SAL group.

4. Discussion

Previously, we demonstrated that BYHWD had a growth promoting activity of peripheral axons after injury (Kim and Namgung, 2018; Chang et al., 2016). Here, we have modified the composition of BYHWD by omitting Lumbricus and adding three herbal drugs AB, PU, and AC that facilitated neurite outgrowth. Given that these drugs were reported to have anti-inflammatory, anti-apoptotic and anti-analgesic activities (Shang et al., 2011; Lee et al., 2019; Xu et al., 2017), a newly formulated ABHD might be more effective than BYHWD for inducing nerve repair and anti-inflammatory reaction, through the pathologic regulation of

injured nerve. Indeed, ABHD downregulated the production of inflammatory cytokines from the injured nerve and DRG neurons, and promoted axonal regeneration and functional recovery after NTC.

Axons in the peripheral nerve can regrow after injury and reinnervate the original target. Yet, the efficiency of functional recovery is, in general, far from perfect and variable, depending on the pathophysiological status of injured nerves. For instance, when the peripheral nerve is surgically reconnected after complete transection, the extent of axonal regeneration leading to functional recovery is lower than the nerve given crush injury, which is partly caused by the reduced reinnervation of regenerating axons toward the original targets (Gordon and Borschel, 2017). In addition, a prolonged denervation after nerve transection can attenuate axonal regeneration, possibly due to a loss of interaction of distal nerve factors in the basal lamina and Schwann cells with regrowing axons (Höke, 2006). Importance of Schwann cell interaction with axons is particularly considered in human case. Since the whole length of peripheral nerve in human is much longer than typical experimental animals such as rats and mice, farther distal portion from the injury site would not be interactive with regenerating axons for a certain amount of time and consequently, may lose the capacity facilitating the regrowth of regenerating axons. By varying the time interval between nerve transection and coaptation, an NTC animal model can be used to investigate molecular factors interacting with regenerating axons (Höke et al., 2002; Gordon, 2016; Kim and Namgung, 2018). Alternatively, accelerated axonal regeneration may overcome the restricted capacity of nerve repair after prolonged denervation (Ma et al., 2011).

We first analyzed the effects of ABHD treatment on axonal regeneration in NTC animals in which the sciatic nerve was transected for 1 week prior to reconnection and found that the distal elongation of regenerating axons was significantly improved by ABHD treatment. Levels of distal extension of regenerating axons was similar between

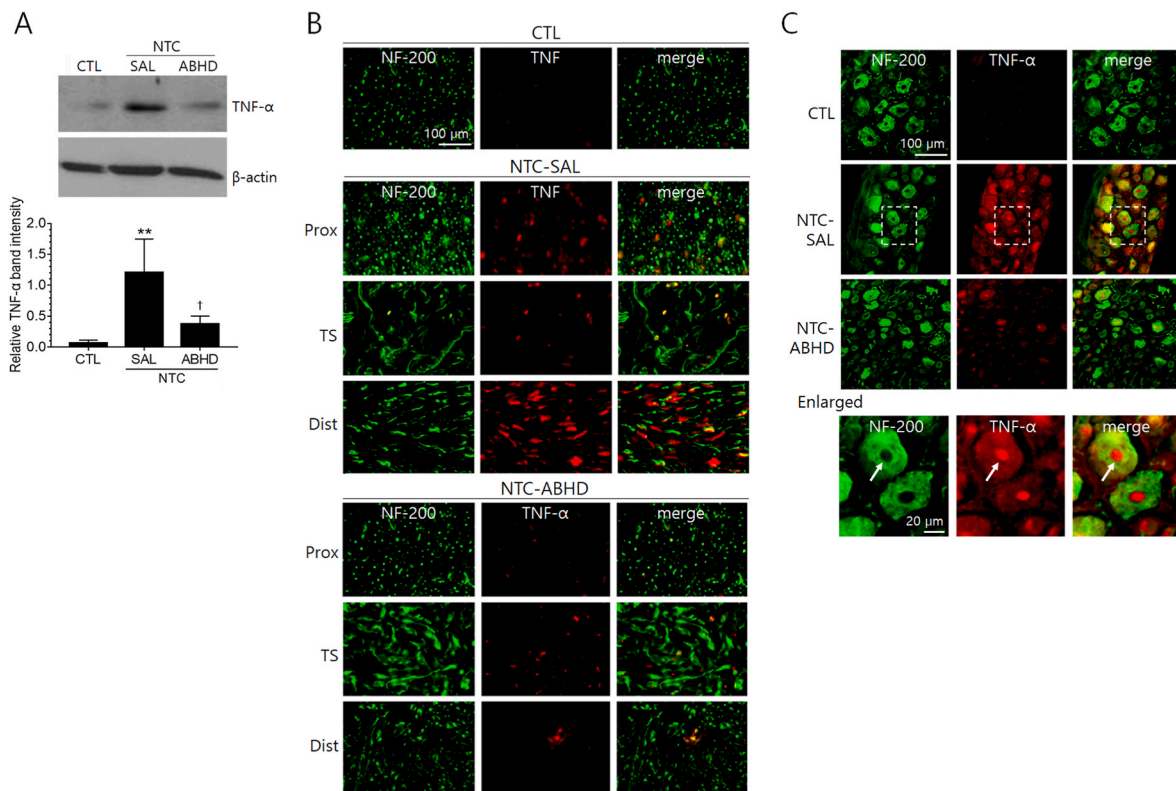


Fig. 5. Effects of ABHD administration on TNF- α production in the sciatic nerve and DRG of NTC animals. (A) Western blot analysis of TNF- α in the sciatic nerve for control (CTL), NTC + SAL, and NTC + ABHD groups of animals. Western blot image (upper) is the representative from four independent experiments, and a quantification of protein band intensity relative to actin is plotted in lower panel. ** $p < 0.01$ vs. CTL; † $p < 0.05$ vs. NTC + SAL. Western blotting for actin as sample loading control. (B, C) Immunofluorescence staining of NF-200 and TNF- α proteins in the sciatic nerve (B) and DRG (C) sections. In (B), transverse sections from the proximal (Prox), transectional (TS), and distal (Dist) portions of the nerve in NTC + SAL and NTC + ABHD groups of animals were used for immunofluorescence staining. In (C), an area of dotted rectangle in NRC + SAL group was enlarged (lowest panel), illustrating the central localization of TNF- α -labeled neurons with surrounding NF-200-stained nuclei. An area of intense TNF- α signals was colocalized with the central zone showing weak NF-200 signals (marked with arrows).

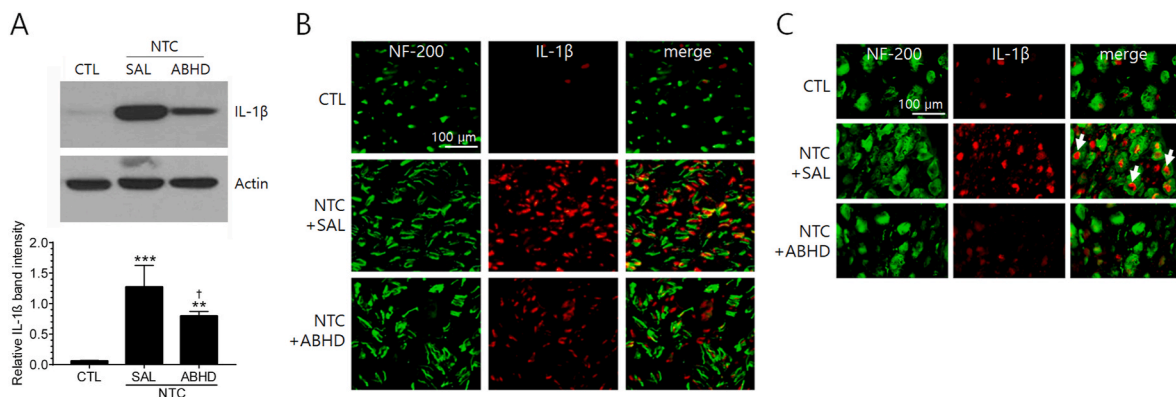


Fig. 6. Effects of ABHD administration on IL-1 β production in the sciatic nerve and DRG of NTC animals. (A) Western blotting of IL-1 β in the sciatic nerve for control (CTL), NTC + SAL, and NTC + ABHD groups of animals. Western blot image (upper) is the representative from four independent experiments, and a quantification of protein band intensity relative to actin is plotted in lower panel. *** $p < 0.01$, *** $p < 0.001$ vs. CTL; † $p < 0.05$ vs. NTC + SAL. Western blotting for actin as sample loading control. (B, C) Immunofluorescence staining of IL-1 β and NF-200 in the transverse sciatic nerves (B) and in the DRG (C) sections.

saline- and ABHD-treated groups near the area of coaptation, but farther elongation was much improved by ABHD treatment. When analyzed 3 months after nerve transection, axonal regeneration through the reconnected distal nerve was much retarded compared with animals with 1 week of transection. Of note is the remarkable improvement of axonal regeneration by ABHD treatment in 3M-NTC animals. Since the nerve was disconnected for 3 months, Schwann cells in the distal stump could be less responsive to regrowing axons than those in acutely disconnected nerve. Accordingly, we speculate that ABHD

administration in 3M-NTC animals may activate Schwann cells in the distal nerve and facilitate their interaction with regenerating axons.

Having found that ABHD was effective at facilitating axonal regeneration, we began to investigate molecular factors that might be involved in the process of axonal regeneration either by acting directly on regenerating axon as an intrinsic factor or by acting indirectly on Schwann cells, an environmental factor. GAP-43 is an axonal growth-associated protein that is strongly induced from neuronal cells after axonal injury and thus has been used as a marker of neuronal capacity of

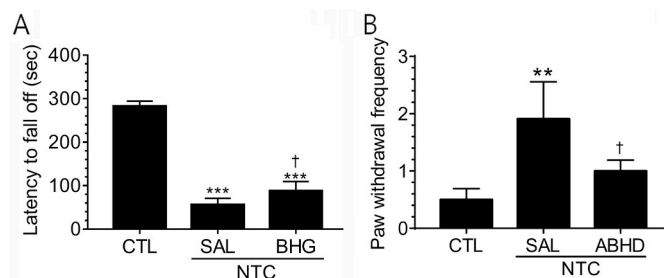


Fig. 7. Behavioral tests to measure the changes in sensorimotor sensitivity after BGH treatment in NTC animals. (A) Rota-rod test and (B) von Frey test. ** $p < 0.01$ and *** $p < 0.001$ vs. untreated control (CTL); † $p < 0.05$ and †† $p < 0.001$ vs. NTC + SAL (One-way ANOVA, number of animals per group = 4).

axonal regeneration (Skene, 1989; Bomzi et al., 2001). Cdc2 is a cyclin-dependent kinase that promotes the transition of G2 to M phase in the cell cycle. Cdc2 is also strongly induced after peripheral nerve injury, but selectively expressed from Schwann cells indicating its role in axonal regeneration as an environmental factor (Han et al., 2007; Namgung, 2014). Neurite length and GAP-43 signals were increased in DRG neurons prepared from rats given preconditioning injury compared with untreated control animals, as has been well demonstrated previously (Smith and Skene, 1997; Liu and Snider, 2001). Levels of GAP-43 induced from injured sciatic nerve were slightly elevated by ABHD but failed to show statistical difference (Fig. 3A). In cultured neurons, GAP-43 signal was detected in both SAL and ABHD groups, showing more intense signals locally at the neuritic tips of ABHD group, and the neurite length was slightly increased by ABHD treatment without showing statistical difference (Fig. 3C). It seems that three new herbal components of ABHD, AB, PU and AC were more effective at inducing neurite growth when treated individually rather than when treated as the components of ABHD, raising the possibility of compositional interaction among ABHD constituents that may interfere the effects. Further study is required to resolve this issue. In contrast, Cdc2 level was upregulated in the injured nerve after ABHD treatment. We have demonstrated that neurite outgrowth of DRG neurons was enhanced by co-culture with Schwann cells expressing Cdc2 (Chang et al., 2012a). Thus, ABHD-mediated increase of Cdc2 production may facilitate distal migration of Schwann cells as well as their proliferation through the reconnected nerves (Han et al., 2007) and lead to enhanced axonal regeneration.

Since we found that ABHD is involved in facilitating axonal regeneration in NTC animals, we further explored molecular factors that might mediate the interaction between non-neuronal cells such as Schwann cells and regenerating axons at the injury site. BDNF-TrkB pathway is known to mediate to transmit the signal between Schwann cells and neurons after nerve injury or electrical stimulation (Luo et al., 2014; Funakoshi et al., 1993; Meyer et al., 1992; Zhang et al., 2000). Here, the production of BDNF was clearly induced in the sciatic nerve after NTC. Since ABHD further upregulated BDNF level, it is likely that BDNF signaling to its axonal receptor may be promoted by ABHD. However, the levels of TrkB, a high affinity receptor of BDNF, were not altered by NTC and ABHD. Instead, phospho-Erk1/2, known to play a role in retrogradely transporting BDNF-TrkB entrapped in a vesicle (Rishal and Fainzilber, 2014; Hanz and Fainzilber, 2006), was colocalized within the axons and its level was elevated by ABHD in the injured nerve and DRG. Consistent with this observation, TrkB protein level was elevated in the DRG sensory neurons of NTC-ABHD animals. Taken together, our data suggest that ABHD treatment increases the production of BDNF from Schwann cells, facilitates retrograde signaling of TrkB with the involvement of Erk1/2 activity, and induces neuronal cell body responses including target gene expression for nerve repair. However, our data do not rule out the possibility that the autocrine interaction of axonally produced BDNF with TrkB receptors may be

induced by ABHD treatment and contribute to enhanced axonal regeneration (Wilhelm et al., 2012).

We have formulated a new herbal decoction in which newly selected constituents AB, PU, and AC were added to the modified BYHWD. Since these herbal components were proposed to regulate antiinflammatory and anti-nociceptive activities (Park et al., 2005; Shang et al., 2011; Lee et al., 2019; Xu et al., 2017), we evaluated whether ABHD was involved in the regulation of the production of major pro-inflammatory cytokines TNF- α and IL-1 β . Immunofluorescence images show that both TNF- α and IL-1 β induced from the sciatic nerve after NTC appeared to be produced from non-neuronal cells around axons, presumably activated Schwann cells and infiltrated macrophages (Shamash et al., 2002; Ruohonen et al., 2005). Our data further showed that production of TNF- α and IL-1 β was also strongly induced from DRG sensory neurons, suggesting that the lesion signal triggered by nerve transection may be retrogradely transmitted and induce soma response including the induction of gene expression of inflammatory cytokines. Protein signals of TNF- α were seen to be particularly strong in the nucleus. Although previous studies have shown nuclear localization of inflammatory cytokines such as TNF- α and IL-1 β , their function in the nucleus is not clear yet, despite the implications on nuclear function (e.g., gene expression and mRNA transport) (Panagakos and Kumar, 1994; Curtis et al., 1990). Future studies on nuclear TNF- α will be of great importance to disclose its function particularly associated with regenerative processes of sensory neurons.

Inflammatory cytokines synthesized and released from neuronal and non-neuronal cells in the injured nerve area may activate Schwann cells and macrophages that subsequently carry out pathological functions such as clearing cell debris and toxic wastes and activation of immune cells (Ide, 1996; Chen et al., 2015). While inflammatory cytokines are advantageous for the repair of injured tissue (Temporin et al., 2008), their prolonged activity could deteriorate the tissue. When we analyzed nerve tissues in 1W-NTC animals, TNF- α and IL-1 β levels, both of which were induced from the sciatic nerve and DRG of NTC animals, were downregulated by ABHD treatment. Although it is unclear at this moment how ABHD downregulated inflammatory cytokine production in the injured nerve and DRG, its function for phenotypic changes of Schwann cells and macrophages can be considered. Distally migrating Schwann cells and anti-inflammatory M2 macrophages following the earlier degeneration stage can increase the production of nerve growth factor and brain-derived neurotrophic factor (BDNF) while decreasing the production of inflammatory cytokines (Kuo et al., 2011; Chen et al., 2015). It was further reported that some herbal formula exert anti-inflammatory effects in macrophages and protect Schwann cells from oxidative injury and promotes the proliferation and migration of Schwann cells (Luo et al., 2020; Cheng et al., 2014; Yang et al., 2011; Fang et al., 2009). Consequently, inflammation-dampening effects of ABHD in NTC animals may contribute to enhanced axonal regeneration and functional recovery.

Our data on behavioral tests showed that the administration of ABHD in NTC animals appears to be effective at improving both sensory and motor activities in affected target organs. In a rotarod test, motor activity was largely weakened in NTC group and improved by ABHD treatment. It was however noted that the level of increases in motor activity by ABHD was about 30% to the control, suggesting a partial recovery of motor function. Since motor neuronal cell bodies are located in the dorsal horn of the spinal cord, the regenerative process of injured axons may take longer than sensory axons. von Frey test showed increases in pain sensitivities in NTC animals, which may reflect hyperalgesic responses mediated by inflammatory cytokines such as TNF- α and IL-1 β induced from the nerve undergoing axonal regeneration. ABHD treatment significantly attenuated pain sensitivity (48% decrease compared to NTC-SAL group), suggesting that ABHD-mediated downregulation of inflammatory cytokines might decrease inflammatory reaction in the coapted nerve. Further studies determining the relative contribution of conventional herbal components of BYHWD vs. newly

added components to anti-inflammatory and anti-analgesic effects and axonal regeneration would be of great importance to gain insight into mechanistic basis of ABHD on peripheral nerve repair.

In conclusion, we have determined a new formulation of herbal decoction ABHD and found that ABHD was effective at facilitating axonal regeneration possibly by dampening inflammation in NTC animals, thus leading to functional recovery. Growth-promoting activity of ABHD even after prolonged denervation implies that it may be applicable to develop therapeutic strategies for the patients suffering from peripheral nerve injury disorders.

Author contribution

KJK, JYP, and UN developed the idea and designed the research. KJK and JH performed experiments. UN wrote the manuscript. All authors read and approved the final manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2020.113063>.

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